4) Experiment to understand the data preprocessing in Data science

Program:

```
import numpy as np
import pandas as pd
df = pd.DataFrame({
    'Country': ['USA', 'France', 'USA', 'Italy', 'France',
                'USA', 'Italy', np.nan, 'France', 'USA'],
    'Age': [27, 34, 29, np.nan, 43, 36, 31, 47, 28, 40],
    'Salary': [45000, 61000, 52000, 65000, np.nan, 55000,
               70000, 78000, 53000, 63000],
    'Purchased': ['Yes', 'No', 'No', 'Yes', 'No',
                  'Yes', 'Yes', 'No', 'Yes', 'No']
})
print("Original Dataset:")
print(df)
print("\n\nDataFrame Information:")
print(df.info())
print("\n\nCountry Mode:")
print(df.Country.mode())
print("\n\nFirst Mode Value:")
print(df.Country.mode()[0])
df['Country'] = df['Country'].fillna(df.Country.mode()[0])
df['Age'] = df['Age'].fillna(df.Age.median())
df['Salary'] = df['Salary'].fillna(round(df.Salary.mean()))
print("\n\nDataset After Filling Missing Values:")
print(df)
print("\n\nOne Hot Encoding:")
print(pd.get_dummies(df.Country))
```

```
updated_dataset = pd.concat(
    [pd.get_dummies(df.Country), df.iloc[:, [1, 2, 3]]],
    axis=1
)
print("\n\nUpdated Dataset:")
print(updated_dataset)
print("\n\nDataFrame Information After Preprocessing:")
print(df.info())
updated_dataset['Purchased'] = updated_dataset['Purchased'].replace(
    ['No', 'Yes'], [0, 1]
)
print("\n\nFinal Dataset:")
print(updated_dataset)
```

Ouput:

```
Original Dataset:
  Country   Age   Salary Purchased
0     USA  27.0  45000.0       Yes
1  France  34.0  61000.0        No
2     USA  29.0  52000.0        No
3   Italy   NaN  65000.0       Yes
4  France  43.0      NaN        No
5     USA  36.0  55000.0       Yes
6   Italy  31.0  70000.0       Yes
7     NaN  47.0  78000.0        No
8  France  28.0  53000.0       Yes
9     USA  40.0  63000.0        No
```

```
DataFrame Information:
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   Country    9 non-null      str
 1   Age        9 non-null      float64
 2   Salary     9 non-null      float64
 3   Purchased  10 non-null     str
dtypes: float64(2), str(2)
memory usage: 312.0 bytes
None
```

```
Country Mode:
0    USA
Name: Country, dtype: str

First Mode Value:
USA
```

Dataset After Filling Missing Values:

	Country	Age	Salary	Purchased
0	USA	27.0	45000.0	Yes
1	France	34.0	61000.0	No
2	USA	29.0	52000.0	No
3	Italy	34.0	65000.0	Yes
4	France	43.0	60222.0	No
5	USA	36.0	55000.0	Yes
6	Italy	31.0	70000.0	Yes
7	USA	47.0	78000.0	No
8	France	28.0	53000.0	Yes
9	USA	40.0	63000.0	No

One Hot Encoding:

	France	Italy	USA
0	False	False	True
1	True	False	False
2	False	False	True
3	False	True	False
4	True	False	False
5	False	False	True
6	False	True	False
7	False	False	True
8	True	False	False
9	False	False	True

Updated Dataset:

	France	Italy	USA	Age	Salary	Purchased
0	False	False	True	27.0	45000.0	Yes
1	True	False	False	34.0	61000.0	No
2	False	False	True	29.0	52000.0	No
3	False	True	False	34.0	65000.0	Yes
4	True	False	False	43.0	60222.0	No
5	False	False	True	36.0	55000.0	Yes
6	False	True	False	31.0	70000.0	Yes
7	False	False	True	47.0	78000.0	No
8	True	False	False	28.0	53000.0	Yes
9	False	False	True	40.0	63000.0	No

```
DataFrame Information After Preprocessing:
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   Country    10 non-null     str
 1   Age        10 non-null     float64
 2   Salary     10 non-null     float64
 3   Purchased  10 non-null     str
dtypes: float64(2), str(2)
memory usage: 312.0 bytes
None
```

Final Dataset:

	France	Italy	USA	Age	Salary	Purchased
0	False	False	True	27.0	45000.0	1
1	True	False	False	34.0	61000.0	0
2	False	False	True	29.0	52000.0	0
3	False	True	False	34.0	65000.0	1
4	True	False	False	43.0	60222.0	0
5	False	False	True	36.0	55000.0	1
6	False	True	False	31.0	70000.0	1
7	False	False	True	47.0	78000.0	0
8	True	False	False	28.0	53000.0	1
9	False	False	True	40.0	63000.0	0

Result:

The data preprocessing in the Data Science is understood.